

PATHIVADA RANJITH KUMAR

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PROFILE / CAREER OBJECTIVE

Computer Science undergraduate (CGPA: 9.3/10) with hands-on experience in Prompt Engineering, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and multi-agent AI systems. Skilled in designing and deploying production-ready AI applications using Python, FastAPI, LangChain, LangGraph, PyTorch, and vector databases. Strong foundation in NLP, Deep Learning, and AI reasoning workflows, with experience building scalable LLM-powered applications and optimizing AI systems for real-world use.

SKILLS & COMPETENCIES

- **Programming:** Python, Java, C, JavaScript
- **Prompt Engineering & GenAI:** Prompt Engineering, LLMs (GPT, Llama, Qwen, Claude), RAG, AI Agents, Multi-Agent Systems, Agentic AI, Vector Search
- **AI/ML:** NLP, Deep Learning, Transformers, Embeddings, Semantic Search, Computer Vision
- **Frameworks:** LangChain, LangGraph, FastAPI, Hugging Face, CrewAI, Streamlit
- **Vector Databases:** Pinecone, FAISS, ChromaDB
- **Databases:** MongoDB
- **Tools:** Git, GitHub, Docker, Linux, Postman, REST APIs
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS

EDUCATION

Rajiv Gandhi University of Knowledge Technologies, Nuzvid	<i>2023 – Present</i>
Bachelor of Technology (B.Tech) in Computer Science	CGPA: 9.3/10
Rajiv Gandhi University of Knowledge Technologies, Nuzvid	<i>2021 – 2023</i>
Pre-University Course (MPC)	CGPA: 10/10

PROJECTS

- AI-Powered Interview Preparation Assistant — Multi-Agent LLM System** github.com/620593/interview-prep
- Designed and developed a LangGraph-based multi-agent LLM system that generates personalized interview preparation roadmaps based on the target company, role, and preparation timeline.
 - Engineered specialized AI agents with role-specific prompts for information extraction, company research, curriculum planning, coding pattern analysis, and study scheduling.
 - Built a FastAPI backend and interactive dashboard to orchestrate AI workflows, generate structured outputs, and track interview preparation progress.
 - Integrated retrieval-augmented generation (RAG) and prompt orchestration to generate personalized interview plans based on company, role, and preparation timeline.
- TriGuard AI — Multimodal Medical Triage Assistant (LangGraph + LLMs)** github.com/620593/TriGuard-AI-Medical-Triage-Assistant
- Developed a production-style multimodal medical triage platform integrating text, voice, OCR, X-ray, and skin image analysis using FastAPI, LangGraph, Groq LLMs, Whisper, Tesseract OCR, and React.
 - Designed an LLM-as-a-Judge validation pipeline where a secondary LLM evaluated structured JSON outputs and clinical reasoning generated by the primary LLM, improving response consistency and reducing hallucinations before final response generation.
 - Implemented secure authentication (Google OAuth + JWT), MongoDB-backed conversational memory, and modular REST APIs, reducing average response latency by **30%** through workflow optimization.
 - Implemented structured JSON outputs and an LLM-as-a-Judge evaluation pipeline to validate responses before presenting them to users.

Handwritten Text Recognition using CRNN (CNN + BiLSTM + CTC)

github.com/620593/

- Designed and implemented a CRNN-based handwritten text recognition model combining CNN feature extraction, BiLSTM sequence modeling, and CTC loss.
- Improved recognition accuracy through transfer learning on synthetic datasets followed by fine-tuning on the IAM dataset, achieving **92% CCR**.
- Developed a complete OCR pipeline for line-level handwritten text recognition without requiring character segmentation.

Content-Based Image Retrieval (CBIR) Studio — ResNet-50, FAISS, Streamlit github.com/620593/Content-Based-Image-Retrieval-CBIR-

- Engineered a Content-Based Image Retrieval (CBIR) system using pretrained ResNet-50 feature embeddings for semantic image similarity search.
- Implemented FAISS IndexFlatL2 vector indexing to retrieve visually similar images from a dataset of approximately 25K images with low-latency nearest-neighbor search.
- Built an end-to-end retrieval pipeline integrating feature extraction, vector indexing, metadata mapping, and an interactive Streamlit interface for real-time image search.

LEADERSHIP & ACTIVITIES

Event Coordinator — Teckzite 2K25, RGUKT Nuzvid

Dec 2025

- Developed an event management dashboard and coordinated the execution of 8+ technical events, collaborating with multiple teams to streamline event operations.

Web Designer — CYGNUS 2K25, RGUKT Nuzvid

Mar 2025

- Designed and maintained the official event website, ensuring a responsive and user-friendly interface for participants.

RELEVANT COURSEWORK

Artificial Intelligence • Machine Learning • Deep Learning • Natural Language Processing • Computer Vision • Data Structures Algorithms • Operating Systems • DBMS

ACHIEVEMENTS

- **NPTEL Artificial Intelligence** — Scored in Top 5% nationally among all participants.
- **NPTEL Cloud Computing** — Ranked in Top 2% nationally among all participants.
- **NPTEL Deep Learning** — Elite + Silver Certificate
- **GATE (2026)** — Score: 697 (AIR 1202).